## Calcolo del ritardo normalizzato

$$\textbf{Ritardo modale } (\textbf{graded} - \textbf{index}) \qquad \frac{\Delta t}{L} = \frac{NA^4}{32 n_{core}^3 c\ [^m/_s]} = \frac{(n_{core}^2 - n_{cladding}^2)^2}{32 \cdot 3 \cdot 10^8 \cdot n_{core}^3}\ \left[\frac{s}{m}\right]$$

$$\textbf{Ritardo da dispersione cromatica} \quad \frac{\Delta t}{L} = \beta_2\ \left[\frac{ps^2}{km}\right] \cdot \Delta\omega\ \left[\frac{rad}{s}\right] = \beta_2 \cdot 2\pi \cdot \Delta f \cdot 10^x\ \left[\frac{s}{m}\right]$$

$$\left[\frac{ps^2}{km}\right] = 10^{-24} \cdot 10^{-3}\ \left[\frac{s^2}{m}\right] = 10^{-24}\ \left[\frac{s^2}{km}\right]$$

$$\textbf{Ritardo da dispersione cromatica} \quad \frac{\Delta t}{L} = D_\lambda\ \left[\frac{ps}{nm \cdot km}\right] \cdot \Delta\lambda\ [nm] =$$

$$= D_\lambda \cdot \Delta\lambda\ \left[\frac{ps}{km}\right] \ = D_\lambda \cdot \Delta\lambda \cdot 10^{-15}\ \left[\frac{s}{m}\right]$$

## Calcolo del prodotto $B_{eff} \cdot L$

$$B_{eff} \cdot L = \frac{0.44}{\frac{\Delta t}{L}\ \left[\frac{\boldsymbol{s}}{\boldsymbol{m}}\right]} = x\ [Hz \cdot m] = x \cdot 10^{-12}\ [GHz \cdot km] = x \cdot 10^{-9}\ [MHz \cdot km]$$

$$B_{eff} \cdot L = \frac{0.44}{\frac{\Delta t}{L}\ \left[\frac{\boldsymbol{ns}}{\boldsymbol{km}}\right]} = x\ [GHz \cdot km] = x \cdot 10^{3}\ [MHz \cdot km]$$

$$B_{eff} \cdot L = \frac{0.44}{\frac{\Delta t}{L}\ \left[\frac{\boldsymbol{ps}}{\boldsymbol{km}}\right]} = x\ [THz \cdot km] = x \cdot 10^{3}\ [GHz \cdot km] = x \cdot 10^{6}\ [MHz \cdot km]$$

## Calcolo della banda efficace del sistema

$$B_{eff} = \frac{B_{eff} \cdot L}{L_{fibra}}\ [Hz]$$

## Calcolo del ritardo da dispersione (modale/cromatica)

$$\Delta t = \frac{\Delta t}{L} \cdot L_{fibra}\ [s]$$